

Behind Cloaks and Masks: Uncovering Deceptive Misuse of Meta’s Advertisement Systems

Anonymous submission

Abstract

Techniques to deceive internet users into accessing malicious content have been employed since the Web 1.0 era, evolving alongside its different iterations. Nowadays, the consolidation of modern Online Social Networks has led to the emergence of new forms of

Introduction

Deceptive practices have been a documented problem and research topic on Web Studies since the Web 1.0 era (Wu and Davison 2006)

RQ1 What forms of URL cloaking can be detected internally within the Ad Library? Which mechanisms are exploited and how prevalent are they?

RQ2 What is the distribution of domains embedded in adverts? When requested, what kinds of redirection cloaking can be observed?

RQ3 What features further differ cloaked and non-cloaked adverts?

This studies contributions to literature are:

- A first of its kind analysis of advert-embedded URLs and cloaking patterns employed in Meta’s Advertising Ecosystem, leading to the uncovering of unintended, malicious abuse of certain Ad Distribution features with the goal of avoiding moderation and deceiving consumers.
- A unified definition of cloaking over more than 20 years of dedicated research, leading to a highlighting of enduring, ressurging and novel deceitful practices and exposing their relation to adjacent research areas.
- A discussion on low-resource moderation strategies that could effectively reduce deceitful patterns.

Related Work

Advertising Ecosystem Auditing

The rise of transparency requirements for digital advertising has led to efforts in measuring how effectively companies are in following this goal (Pochat et al. 2022). Auditing becomes the object of study itself, with policies compared between platforms, regulations and public opinion (Sosnovik and Goga 2021). A study focusing on Facebook has shown

how algorithmic advert delivery optimization can lead to discriminatory content distribution (Ali et al. 2019).

The abuse of otherwise legitimate distribution systems for malicious purposes has been a research topic in several studies. In (Papadogiannakis et al. 2025), authors investigate how identifier pooling can be used to redirect revenue to fraudulent sources. Through the analysis of four different traffic sources, a study by (Szurdi et al. 2021) investigates abusive advertising distribution, revealing that cloaking-aware scraping leads to the discovery of 81% more malicious URLs than singular crawling strategies, also highlighting the relevance of URL shorteners in fraudulent advertising.

Meta’s Ad Library has been extensively studied in the context of political advertising (Aisenpreis, Gyrst, and Sekara 2023; Civelli et al. 2026; Islam, Roy, and Goldwasser 2023; Inacio da Silva et al. 2026), with the ecosystem also being investigated externally (Silva et al. 2020; Gkiouzepe et al. 2023) from the library to further transparency efforts. It has also been studied in the context of Illicit Betting Ecosystems (Vaidya and Garimella 2026), Health-related topics, such as “alternative” medicine promotion (Zenone et al. 2023) and COVID-19 informational weaponization (Silva and Benevenuto 2021). One security-centered study points to design and implementation flaws that harm the Library’s transparency, revealing the presence of undeclared, coordinated advertising campaigns associated with inauthentic content (Edelson, Lauinger, and McCoy 2020).

Cloaking

With over 20 years both as a conceptual research topic and as a set of strategies for online deception, the term’s usage has evolved with the different, emerging strategies that enable cloaking in different contexts.

Redirection Cloaking: Research by (Zeng et al. 2022) provides a general understanding of redirection mechanisms used in cloaking, outlining the same practices we empirically observe.

Url Masking: A study by (Fukushi et al. 2021) analyzed the security risks of ad-based URL shortening services.

Creative Optimization: The exploitation of dynamic content delivery systems avoid moderation has been a phenomenon reported in a wide range of platforms, being confirmed as an effect bypass for contextual ads in Youtube

(El Fraihi, Abdelaziz, and Goga 2026)

Currently, the closest a study has gotten to analyzing cloaking patterns in an online ecosystem is (Bartuzi-Trokielewicz, Martinek, and Trokielewicz 2026), which does not mention the specific platform.

Research Gap

Despite plentiful related analyses, research has frequently focused on context-specific definitions (Szurdi et al. 2021), leading to rich, highly diverse reporting on actively employed cloaking techniques. However, such diversity has also led to similar phenomena not being considered under an unifying framework.

With, to the best of our knowledge, no previous studies on advert-embedded URLs on Meta’s advertising ecosystem, our study aims to fill in:

1. A conceptual gap on the definition of cloaking through an analysis of its occurrence on in one of the most relevant advertising ecosystems in the world.
2. A gap in security auditing literature where no large-scale analyses have been conducted in relation to URL embeds for adverts in Meta’s Ad Library.

Methodology

Data Collection

Meta provides access to complete advertising information through its Web interface, with API access in most countries covering only ads about politics and social issues¹. To ensure data are as complete as possible, we thus focused our retrieval efforts entirely the Web platform, through a collection pipeline whose closest literature match is (Pochat et al. 2022), with our key differences including complete multimedia creative extraction (images and video) and increased diversity coverage through an unfiltered search approach instead of a advertiser-constrained one. Our crawler was operated for two months, from the 20th of April to the 19th of June 2026, resulting in the collection of 407771 full advert media + metadata pairs, taking over more than 600GB storage space.

In order to provide a thorough analysis of Meta’s Ad Library, our collection pipeline was implemented prioritizing the advert’s data completeness and granularity over higher-scale collection of partial content. For instance, although an advert’s multimedia content is its most prominently user-facing component, it is also the most difficult one to work with from a data collection standpoint, with images and especially video taking more storage space and bandwidth than JSON-based GraphQL query results while also adding in complexity regarding file formats and binary storage. This factors into the reasons why most studies only partially include or entirely discard multimedia content.

A metadata-only study allowed researchers to create and analyze a dataset of over 3 million political adverts in circulation in the US (Edelson, Lauinger, and McCoy 2020). It dedicates an entire section to un retrievable adverts, including a bug report and subsequent fix by Meta’s teams,

which reduced but did not completely solve the issue. The study also mentions how, at the time, access to ads was provided only through search queries or advertiser identifiers, the latter being the basis for their advertiser-centric collection. Their approach works well in the political context, where the landscape of advertisers is constrained within a defined thematic boundary, making possible information loss from not including emerging new advertisers not as relevant. This approach, however, does not work well for our purposes, as constraining the queries to specific sets of advertisers by definition adds in bias particularly harmful to the investigation of moderation avoidance techniques, as *a priori* mapping of advertisers associated with cloaking would. Our approach solves both the biased and inconsistent advert retrieval issues through the use of special characters as the search term. We’ve empirically verified that searches for diacritics and characters like ‘*’ return the content expected for a generic search in the following scenarios:

- Searches for ‘keyword *’ return the same view as ‘keyword’. Positioning of the special character does not alter this result.
- Searches for ‘*’ under a specific advertiser an identical view than the full advert display
- Searches for ‘*’ in countries with ≥50k adverts (the maximum reported ad amount) return ≥ than vowel searches return.
- Searches for ‘*’ in advertisers
- Searches for ‘`’, ‘~’, ‘^’ and other special characters return identical results in the previous scenarios.

We therefore operate under the assumption that in searches with no other parameters, the return value is equivalent to an unfiltered search. This assumption is strengthened by the diversity of advertised content in the dataset and by confirming that this technique allows for the collection of adverts that are neither displayed nor retrievable on their respective advertiser pages, highlighting the increased granularity. We consider these conditions to be sufficient for operating a novel analysis that makes the most of noted limitations and inconsistencies found in the underdocumented Ad Library Ecosystem (Gkiouzepi et al. 2023; Pochat et al. 2022; Vaidya and Garimella 2026; Edelson, Lauinger, and McCoy 2020).

One study including media collection (Vaidya and Garimella 2026) analyzed Meta’s adverts as a component in the context of the illegal betting ecosystem in India, with a dataset containing 29.772 ads of which 23.620 had their media content retrieved, attributing it “to missing or dynamically loaded elements”. In order to further the goal of thoroughness, we launched preliminary metadata-only collection steps to map advert display formats, ensuring our data ingestion component could handle all detected dynamic creatives. Table 1 shows all detected formats, with ‘VIDEO’, ‘IMAGE’, ‘DCO’ and ‘CAROUSEL’ dominating over the remaining categories. Surprisingly, the reported total amount of images and video successfully downloaded in this betting study was about 79.3%, close to the proportion of 81.1% we find adding in the totals for ‘VIDEO’ and ‘IMAGE’ formats in the table, although images are the most

¹<https://www.facebook.com/ads/library/api>

Display Format	Total	URL Masking	Card Cloaking	Multi Redirection	Single Redirection	Total Cloaked
VIDEO	223002 (54.6%)	11505 (5.2%)	-	5664 (2.5%)	5493 (2.5%)	16863 (7.6%)
IMAGE	112877 (27.7%)	5944 (5.3%)	-	2202 (2.0%)	908 (0.8%)	6744 (6.0%)
MULTI.IMAGES	3153 (<0.1%)	0	-	0	3 (0.1%)	3 (0.1%)
MULTI.VIDEOS	143 (<0.1%)	0	-	0	0	0
MULTI.MEDIAS	39 (<0.1%)	0	-	0	0	0
DCO	38846 (9.5%)	1768 (4.6%)	883 (2.3%)	440 (1.1%)	375 (1.0%)	2336 (6.0%)
CAROUSEL	27032 (6.6%)	120 (0.4%)	44 (0.2%)	11 (<0.1%)	162 (0.6%)	308 (1.1%)
DPA	2453 (<0.1%)	132 (5.4%)	2 (0.1%)	3 (0.1%)	543 (22.1%)	658 (26.8%)
PAGE.LIKE	141 (<0.1%)	0	-	2 (1.4%)	5 (3.5%)	7 (5.0%)
TEXT	71 (<0.1%)	1 (1.4%)	-	0	1 (1.4%)	2 (2.8%)
EVENT	14 (<0.1%)	0	-	0	0	0
Dataset	407771	19470	929	8322	7490	26921 (6.6%)

Table 1: Total collection amounts.

frequent media type found in India while our Brazil-based dataset has video as the most popular format. In the present study, we find the dynamic advert (DCO, Carousel and DPA) proportion (16.6%) to also be in matching range of the reported 20.6% of ads with media unsuccessfully extracted “due to missing or dynamically loaded content and access instability”.

Cloaking Detection

URL Masking URL Masking happens through the use of the field “Caption”, which essentially works as anchor text, allowing for hyperlinks to be shown as arbitrary strings in place of the actually embedded link. Figure 3 highlights how the difference between “caption” and “link_url” misleads the audience into believing they are accessing a website different from what’s actually linked. This type of misuse of flexible hyperlinking has been reported as associated with phishing (Odgerel et al. 2025; Zhang et al. 2021) and increased user susceptibility to it (Zhuo et al. 2024b,a).

URL Masking detection is implemented through the comparison of domain names. As captions are sometimes used as text fields, non-URL formatted strings are discarded, followed by domain name comparison. The library was used TLDextract² was used to ensure updated Public Suffix List compatibility and private domain extraction.

Card Masking Card masking is detected when one link_url differs from the others, such as in an advert where 4 cards embed an actual “amazon.com” link while one has a link to “othersite.xyz”.

URL Resolution Cloaking URLs were requested from 4 differently geolocated IPs in Brazil, Portugal, Germany and the US, with requisitions repeated 5 times to capture possible variation.

Limitations

In the context of the already established limitations inherent to the Ad Library (Gkiouzepi et al. 2023; Pochat et al. 2022; Vaidya and Garimella 2026), the main limitation of this study comes from the fact that we cannot know the total ads in circulation during the collection timeline, .

²<https://pypi.org/project/tldextract/>

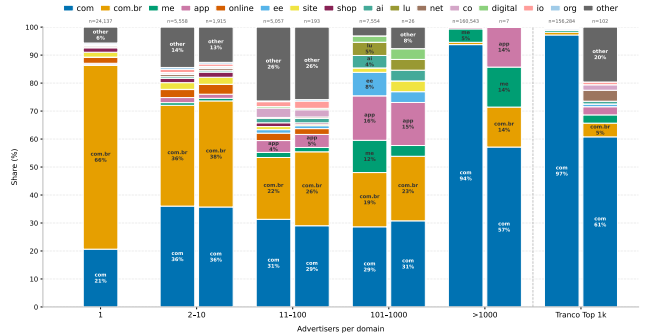


Figure 1: Top-Level Domain Distribution for websites linked by different ranges of total advertisers. Left bars are weighted by advertiser amount, while right bars count only unique domains on the specified range.

Results

Domain Distribution

The total 26279 unique domains found are observed to be distributed with severely skewed frequencies, with the most commonly used ones being dominated by Meta’s own platforms, highlighting the circularity of Meta’s ecosystem. For instance, “whatsapp.com” is used over 50 times more than the most frequent non-Meta domain “shopee.com”. **Domain Usage by Advertiser.**

Internal Cloaking

This section describes deceptive URL patterns that can be detected entirely within Meta’s advertising systems, not requiring external HTTP requests as these patterns are detected entirely in the advert’s metadata JSON.

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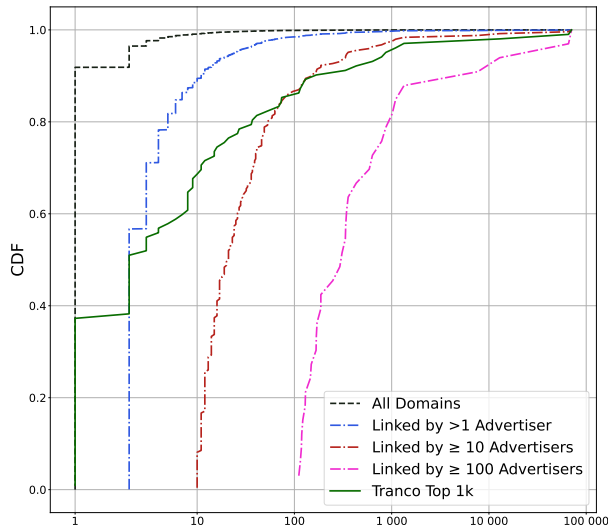


Figure 2: Url Distribution

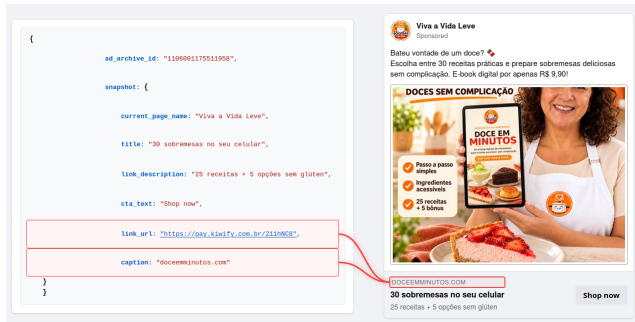


Figure 3: A card-cloaked image advert. It leads the audience into believing the website domain is “doceeminutos.com” while the actually embedded URL is “pay.kiwify.com.br/211hNC8”.

ible hyperlinking has been reported to be associated with phishing susceptibility (Zhuo et al. 2024b).

Card Cloaking Card cloaking happens when Dynamic Advertising formats are abused to hide malicious content amid harmless fillers.

External Cloaking

What features further differ cloaked adverts from non-cloaked ones?

Comparative Analysis

Discussion

Cloaking Detection and User Protection We suggest a complete removal of the url captioning feature, forcing true URLs to be exposed and always leaving it clearly shown to the end user.

We suggest static analysis of dynamic adverts to detected discrepancies in content.

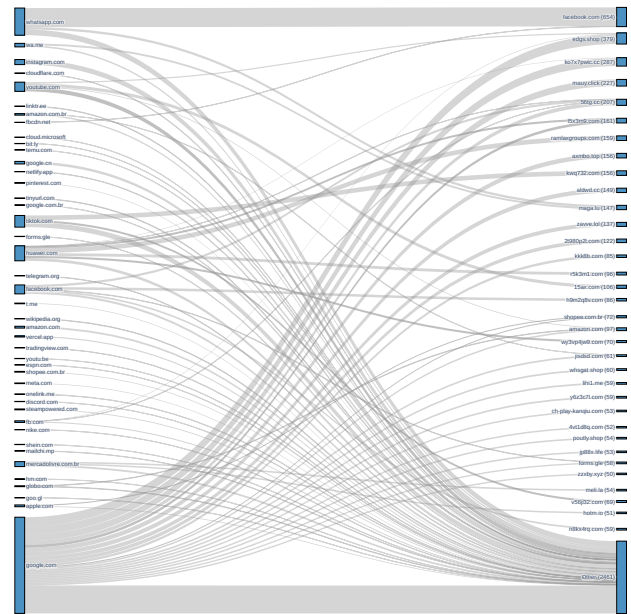


Figure 4: Masks using domains from Tranco (Le Pochat et al. 2019)

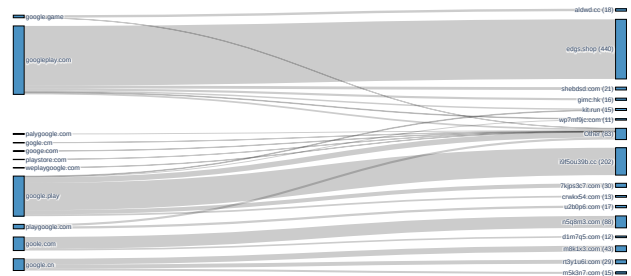


Figure 5: Masks simulating google.

We suggest fingerprinting media to ease the identification of persistent campaigns.

Preparing an Anonymous Submission

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